

STACK

Q8)Hassan Building Jump Stamina (Next Greater Element XOR)

DS	Session	Stack	Question Information	Level 1 Challenge 48
Question description Hassan enjoys jumping from one building to the next. However, he merely jumps to the next higher building and stops when there are none accessible. The amount of stamina necessary for a voyage is equal to the xor of all the heights Hassan leaps till he comes to a halt. If heights are [1 2 4], and he starts from 1, goes to 2 stamina required is $1 \oplus 2 = 3$, then from 2 to 3. Stamina for the entire journey is $1 \oplus 2 \oplus 4 = 7$. Find the maximum stamina required if can start his journey from any building. Constraints $1 \leq N \leq 10^5$ $1 \leq \text{Height} \leq 10^9$ Input First line: N, no of buildings. Second line: N integers, defining heights of buildings. Output Single Integer is the maximum stamina required for any journey. Explanation: 8				

CODE:

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C main.c > main
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main(void) {
5      int n;
6      if (scanf(format: "%d", &n) != 1) return 0;
7
8      long long *h = (long long *)malloc(size: n * sizeof(long long));
9      for (int i = 0; i < n; i++) {
10         scanf(format: "%lld", &h[i]);
11     }
12
13     int *stack = (int *)malloc(size: n * sizeof(int));
14     int top = -1;
15     long long *dp = (long long *)malloc(size: n * sizeof(long long));
16     long long max_stamina = 0;
17
18     for (int i = n - 1; i >= 0; i--) {
19         while (top >= 0 && h[stack[top]] <= h[i]) {
20             top--;
21         }
22
23         if (top == -1) {
24             dp[i] = h[i];
25         } else {
26             dp[i] = h[i] ^ dp[stack[top]];
27         }
28
29         stack[++top] = i;
30
31         if (dp[i] > max_stamina) {
32             max_stamina = dp[i];
33         }
34     }
35
36     printf(format: "%lld\n", max_stamina);
37
38     free(ptr: h);
39     free(ptr: stack);
40     free(ptr: dp);
41     return 0;
42 }

```

OUTPUT:

```
• [suraj@suraj-83gs test]$ ./main
3
1 2 4
7
○ [suraj@suraj-83gs test]$
```

v